

Normalizing Flows

Probability Transport and Invertible Models

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Generative Modeling Problem

We wish to model a probability density:

$$p(x), \quad x \in \mathbb{R}^d$$

Goal:

- tractable likelihood
- efficient sampling
- expressive model

Key Idea

Start with simple distribution:

$$z \sim p_0(z), \quad p_0 = \mathcal{N}(0, I)$$

Apply invertible map:

$$x = f(z)$$

Then:

$$p(x) = p_0(z) \left| \det \frac{\partial z}{\partial x} \right|$$

Change of Variables Theorem

For invertible f :

$$x = f(z)$$

$$p(x) = p_0(z) \left| \det \frac{\partial f^{-1}}{\partial x} \right|$$

Equivalently:

$$p(x) = p_0(z) \left| \det \frac{\partial f}{\partial z} \right|^{-1}$$

$$\log p(x) = \log p_0(z) - \log \left| \det \frac{\partial f}{\partial z} \right|$$

Define:

$$x = f_K \circ f_{K-1} \circ \cdots \circ f_1(z)$$

$$\log p(x) = \log p_0(z) - \sum_{k=1}^K \log \left| \det \frac{\partial f_k}{\partial h_{k-1}} \right|$$

Interpretation

- Each layer transports probability mass
- Total transformation = composition

Affine Transformation

$$x = Az + b$$

$$\det J = \det A$$

$$f(z) = z + u \tanh(w^T z + b)$$

Jacobian:

$$\det J = 1 + u^T \psi(z)$$

$$f(z) = z + \beta \frac{z - z_0}{\alpha + \|z - z_0\|}$$

Split variables:

$$z = (z_1, z_2)$$

Transform:

$$x_1 = z_1, \quad x_2 = z_2 \odot e^{s(z_1)} + t(z_1)$$

$$\det J = \exp \left(\sum s(z_1) \right)$$

- triangular Jacobian $\rightarrow$ efficient

Autoregressive Model

$$x_i = f_i(z_i; x_1, \dots, x_{i-1})$$

$$\det J = \prod_i \frac{\partial x_i}{\partial z_i}$$

$$\frac{dx}{dt} = v(x, t)$$

$$\frac{d}{dt} \log p(x(t)) = -\nabla \cdot v(x, t)$$

Connection to Physics

- continuity equation
- Liouville equation

Flows as Diffeomorphisms

$$f : \mathbb{R}^d \rightarrow \mathbb{R}^d$$

- invertible
- smooth

- probability distributions as densities on manifold
- flow = transport map

$$p(x) = \frac{1}{Z} e^{-E(x)}$$

Partition Function Problem

$$Z = \int e^{-E(x)} dx$$

- hard to compute

$$p(x) = p_0(z) \left| \det \frac{\partial z}{\partial x} \right|$$

- replaces partition function with Jacobian

$$T : p_0(z) \rightarrow p(x)$$

Monge Problem

$$\min_T \int \|x - T(x)\|^2 p_0(x) dx$$

- flows approximate transport maps
- connect to Wasserstein geometry

Maximum Likelihood

$$\mathcal{L} = \sum_i \log p(x_i)$$

$$\nabla_{\theta} \log p(x) = \nabla_{\theta} \log p_0(z) - \nabla_{\theta} \log |\det J|$$

Challenges

- invertibility constraint
- Jacobian computation
- expressivity vs tractability

Summary

- normalizing flows = invertible transformations
- exact likelihood computation
- deep connection to physics and geometry